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Spatial Biodiversity Modeling Pipeline and Benchmark

Caleb Caulk

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Department of Information Technology
Uppsala University
Sweden
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1 Introduction

1.1 Motivation

Biodiversity is defined as "the variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species and of ecosystems."[1] Biodiversity, in short, is the collection of all living organisms of an ecosystem from plants to animals to fungi. Biodiversity is integral to the functions and stability of ecosystems and plays a pivotal role in maintaining ecosystem functions, recycling nutrients, breaking down pollutants, regulating water systems, carbon storage,stabilizing total biomass within a community, and driving the productivity of an ecosystem (Stork 2009; Cardinale et al. 2012; Gascon et al. 2015). Furthermore biodiversity's significant for human survival is immense, as it provides food, materials, medicine, water security, health, well-being, energy, transportation, climate regulation, and more (Stork 2009; Gascon et al. 2015; Brauman et al. 2020; Kim et al. 2024).

Despite the overwhelming benefits of biodiversity to humanity and ecosystems, biodiversity is declining at an unprecedented rate. Out of all the world's terrestrial wilderness ecosystems approximately 74% of this land area has experienced erosion and approximately 9.6% has been completely lost since the early 1990s (Watson et al. 2016). Some of the most impacted areas of biodiversity loss include marine ecosystems where 66% of the ocean's surface has been significantly impacted by human activities (Díaz et al. 2019). Moreover, it is reasonably likely that most of coral reefs will be lost if global temperatures exceed 2°C (Frieler et al. 2012). Furthermore, human activities threaten the survival of approximately 1 million species, 25% of plants and animals are threatened, and human activities have caused an explosion in the rate of species extinction, which is now estimated to be around 1,000 times the natural background rate (Pimm et al. 2014; Díaz et al. 2019). To make matters worse, conservation efforts are underfunded, with a deficit exceeding half a trillion US dollars annually (Deutz et al. 2020). The acceleration of taxonomic loss and the drastic underfunding of conservation efforts have necessitated a shift towards prioritization of biodiversity conservation and ecological rehabilitation.

Many initiatives around the world are being adopted to combat biodiversity loss and climate change which are intricately linked. These initiatives focus on international cooperation, expanding protected areas, reducing pollution, and incentives for sustainable practices (IPBES 2019; IPCC 2023). Specific policy initiatives such as the 30x30 target-which aims to protect 30% of the world's land area by 2030-represent critical efforts for ecological recovery. If successful, we could see large gains and rebounds in biodiversity. Achieving these goals would have significant impacts including: carbon sequestration, increasing water quality regulation, and expanding capabilities to mitigate nutrient pollution (Zeng, Koh, and Wilcove 2022). For these initiatives to be successful, it is crucial to have accurate and robust models that can predict biodiversity patterns and trends. These models can inform conservation strategies, identify priority areas for protection, and assess the potential impacts of different conservation actions.

Species distribution models (SDMs) are some of the most common models for biodiversity prediction. SDMs are a class of models that predict the occurrence (presence or absence) or abundance of individual species based on a wide range of environmental, climate, anthropogenic, and spatial variables (Beery et al. 2021). SDMs have been used extensively for over the past few decades as advances in computational power and data availability have further enhanced their utility and the diversity in modeling approaches (Özkan Tümer et al. 2026; Norberg et al. 2019). These models are powerful for understanding individual species' distributions and ranges, revealing biological patterns, and helping with discovery of sites for the reintroduction of threatened species (Sinclair, White, and Newell 2010). Beyond species-level predictions, community-level or biodiversity metric predictions can be constructed from several SDMs and then aggregating all model results together. This is commonly referred to as the "predict first, assemble later" or bottom-up approach (Ferrier and Guisan 2006). This bottom-up approach, while intuitive, possesses several critical shortcomings: it exhibits high sensitivity to data quality, model selection, and parameter fitting (Araújo and Guisan 2006); it frequently neglects biotic species interactions and lacks the computational efficiency to scale for large communities (Ferrier and Guisan 2006); it struggles with transferability to unsampled sites (A. Lee-Yaw et al. 2022); and it often produces biased estimates at the 'tails' of abundance distributions, over-predicting rare sites and under-predicting high-density areas

[1] Convention on Biological Diversity (1993) *Article 2. Use of terms.* Available at: https://www.cbd.int/convention/articles/?a=cbd-02 (Accessed: 8 April 2026).

(Waldock et al. 2022).

Currently, the most common models for biodiversity prediction aggregates species distribution models (SDMs), which predict individual species' occurrence or abundance. This "predict first then combine" or bottom-up approach is widespread but suffers from key shortcomings: neglecting species interactions, poorly accounting for rare species, and struggling with large numbers of species **spatialmodellingofbiodiversityCLM**, **SDMRarelypredictrealpopulations**. While joint SDMs, which model multiple species at once, offer solutions to these shortcomings, an alternative "combine first predict later" or top-down approach offers benefits, such as handling numerous species simultaneously, better scaling properties, better model calibration through 'averaging' of species-level errors and potentially better at capturing overall diversity at a higher level **spatialmodellingofbiodiversityCLM**. While these top-down methods are more intuitive for modeling biodiversity as a whole, they remain underdeveloped compared to SDMs and lack standardized terminology and methods. Species richness models, macro-ecological models, and generalized dissimilarity models (GDMs) all predict biodiversity but employ different terminology and metrics **GobalmodelsplantdiversityMLtechniques**, **MacroecologicalModelsVsSSDMs**, **GDMGuide**.

1.2 Research Questions

2 Background

This section will cover the background of biodiversity modeling and the different types of models that have been developed for predicting biodiversity patterns and trends. We will start with a general overview of biodiversity and taxa, then move on to species distribution models (SDMs), joint species distribution models (JSDMs), and finally community level modeling approaches. We will discuss the strengths and weaknesses of each modeling family and conclude with a discussion of the need for a standardized pipeline for community level modeling and how this master's thesis aims to fill that gap.

2.1 Biodiversity and Taxa

2.2 Model Families

In this subsection, we will discuss the different model families that have been developed for predicting biodiversity patterns and trends. We will start with species distribution models (SDMs), which are the most common models for biodiversity prediction. We will then move on to joint species distribution models (JSDMs), which model multiple species at once, and finally community level modeling approaches, which model biodiversity as a whole. We will discuss the strengths and weaknesses of each modeling family and how they can be used to predict biodiversity patterns and trends.

2.2.1 Species Distribution Models

Species Distribution Models (SDMs) are a class of models that predict the occurrence (presence or absence) or abundance of individual species at unique geographic locations. SDMs utilize a wide range of variables such as temperature, precipitation, land cover, and human presence to make predictions on the distribution of individual species by establishing relationships between these variables and species occurrences (Beery et al. 2021). While the first SDMs were developed in the early 20th century with distributions of the Chestnut-backed Chickadee (Grinnell 1904), the application of these models has been extended beyond just generating range maps (Beery et al. 2021). They now encompass the pursuit of insights into the causal drivers of species distributions, land use management planning, conservation planning (Elith and Leathwick 2009), revealing biological patterns, and even helping with the discovery of sites for the reintroduction of threatened species (Sinclair, White, and Newell 2010). The versatility of these models has led to their widespread use in ecology and conservation biology especially in the past few decades.

The utility of SDMs was first expanded by the development of the Generalized Linear Model (GLM) and Generalized Additive Model (GAM) frameworks in the late 1970s and 1980s. Before these developments SDMs were primarily ecological models based on conceptual or expert knowledge of species distributions. One of the largest challenges of statistical SDMs models was the assumption of Gaussian error distributions for linear models. This assumption is often violated in ecological data. For example, count data follow a Poisson distribution (Guisan, Edwards, and T. Hastie 2002). The main breakthrough was first with GLMs. GLMs are able to model non-Gaussian-distributed data; specifically, any distribution in the exponential family can be modeled with a GLM (Nelder and Wedderburn 1972). GAMs then further extended the capabilities of GLMs by allowing for nonlinear relationships between the predictors and the response variable (T. J. Hastie and Tibshirani 1986). However, even with the advance of GLMs and GAMs to model non-Gaussian data and nonlinear relationships, SDMs still would struggle when there would be a dependency structure in the data and violated independent and identically distribution (i.i.d.) assumptions d(Guisan, Edwards, and T. Hastie 2002; Beery et al. 2021). To address these issues and the need to model more complex relationships machine learning models including random forests, boosted regression trees, gradient boosting, neural networks, and ensemble methods have been increasingly used (Norberg et al. 2019; Beery et al. 2021; Valavi et al. 2022; Waldock et al. 2022; Özkan Tümer et al. 2026). These methods can capture complex nonlinear relationships and interactions between predictors and the response variable, and they can handle large datasets with many predictor variables.

Concurrently, large advancements in the organization and distribution of biodiversity data through or-

ganizations such as the Global Biodiversity Information Facility (GBIF)[2] and national databases has reduced barriers to data access (Özkan Tümer et al. 2026). In addition to open biodiversity datasets, the quality of biodiversity datasets has continued to improve the quality of species data (Araújo, Anderson, et al. 2019). Moreover, advancements in geographical information systems (GIS) have allowed for the integration of spatial data into SDMs. This spatial data can be of climatic, environmental, topographic, or anthropogenic nature across larger areas and finer temporal scales (Elith and Leathwick 2009; Waltari et al. 2014; Özkan Tümer et al. 2026). These advancements alongside with the development of new modeling techniques have further driven the advancement of SDMs with dozens of models being developed across hundreds of papers in the past few decades (Valavi et al. 2022; Özkan Tümer et al. 2026).

SDMs have been the dominant modeling family for species prediction for several decades and have been widely used in ecology and conservation biology and for good reason. They excel at predicting the distribution of individual species and contribute to our understanding of species niches (Sinclair, White, and Newell 2010). However, they are not without their shortcomings. Firstly, and the most obvious when it comes to measuring biodiversity is that SDMs are designed to predict the distribution of individual species (Ferrier and Guisan 2006), and most SDMs do not capture biotic interactions between species, which can be a critical driver of species distributions and community composition (Waldock et al. 2022). If we want to derive biodiversity metrics from SDMs, we would have to fit a separate SDM for each species and then aggregate the results together in a "predict first, assemble later" or bottom-up approach. This can be in itself can be very time-consuming not only to building the models themselves but also to gather all the species data and environmental data across spatial and temporal scales (Ferrier and Guisan 2006). Compounding this issue is the fact that most species data is still presence only data which would make biodiversity metric calculations aside from species richness difficult (Valavi et al. 2022).

Furthermore, all SDMs are not created equally in varying environmental contexts. In some contexts such as measuring abundance, a random forest may be the best model (Waldock et al. 2022), while in a presence only context an ensemble based model may be the best (Valavi et al. 2022). Other studies have found large variations in model performance and one model may not rule them all and that the best model can vary across species and contexts (Norberg et al. 2019). This is further exacerbated by the fact that hyperparameter tuning or parameterization and feature selection models can also have a large effect on the model performance (Araújo and Guisan 2006; Beery et al. 2021). So, if we wanted to use SDMs to predict biodiversity metrics, not only would we have to fit a separate SDM for each species but it is likely we would have to fit multiple models, and even multiple versions of the same model, for each species to find the best model for each species. SDMs also struggle to predict rare or infrequent species which can account for a large portion of species richness metrics (Ferrier and Guisan 2006; Norberg et al. 2019). Many studies also note how transferability or extrapolation to unsampled sites is a common challenge for SDMs and is in part due to autocorrelation of environmental and spatial variables (Norberg et al. 2019; A. Lee-Yaw et al. 2022). Finally, SDMs tend to be biased at the extremes of abundance distributions, frequently overestimating low-abundance sites while underestimating sites with high abundance (Waldock et al. 2022).

We can see that SDMs have advanced from ecological models based on expert knowledge (Grinnell 1904), to parametric statistical models (Guisan, Edwards, and T. Hastie 2002), to machine learning models (Özkan Tümer et al. 2026). The advancement of SDMs has concurrently been driven by advancements in quality of species data as a response variable and (Araújo, Anderson, et al. 2019), environmental data as predictor variables (Beery et al. 2021). SDMs can do well when it comes to prediction occurrences of individual species (A. Lee-Yaw et al. 2022); however, the shortcomings of SDMs as a model family when it comes to modeling biodiversity metrics are significant. While aggregating multiple SDMs can be used to predict biodiversity metrics, other methods such as joint SDMs (JSMDs) have shown to be more effective at predicting biodiversity metrics (Norberg et al. 2019).

2.3 Joint Species Distribution Models

2.4 Community Level Modeling

[2]Global Biodiversity Information Facility (GBIF). Available at: https://www.gbif.org/ (Accessed: 19 April 2026).

3 Methods

4 Experiments

5 Discussion

6 Conclusion

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